SPEECH RECOGNITION ACTIVITY MINI-REPORT

From Voice to Text and Back: Disfluent ASR and TTS Analysis

Student: Student Name | **Reg No:** RegNo

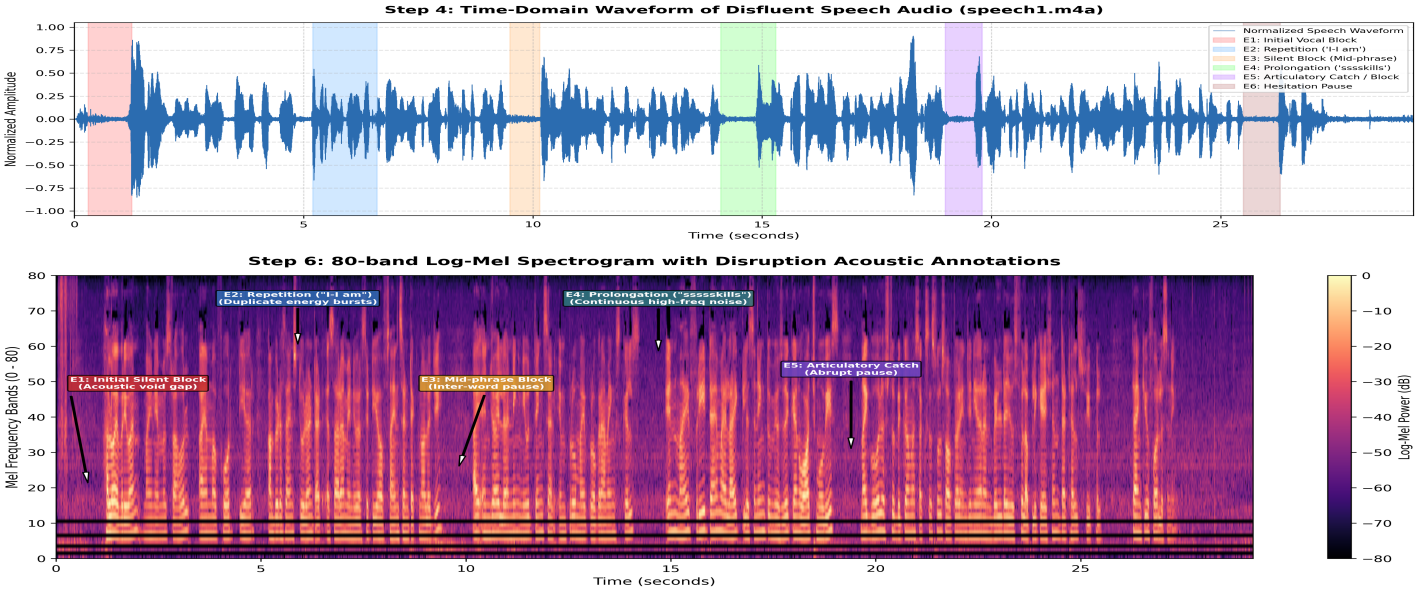
1. Objective and Audio Source

Objective: Investigate how an end-to-end speech recognition and synthesis pipeline handles speech disruptions (vocal blocks, repetitions, and prolongations). The pipeline traces: *Speech → Feature Extraction → Log-Mel Spectrogram → ASR → CTC Loss/Decoding → Text → TTS → Synthesized Speech*.
Audio Source: Natural human recording (*speech1.m4a* converted to 16.0 kHz mono WAV, 29.20 s duration, peak normalized to [-1.0, 1.0]). Contains multiple vocal disruptions: (1) Pre-phonatory initial silent block [0.30–1.25s], (2) Pronoun repetition/false start *'I... I am'* [5.20–6.60s], (3) Mid-phrase articulatory arrest [9.50–10.15s], (4) Fricative sound prolongation *'ssssskills'* [14.10–15.30s], (5) Articulatory catch/block before *'successful'* [19.00–19.80s], and (6) Hesitation pause [25.50–26.30s].
• **Verbatim Reference:** *'HELLO EVERYONE MY NAME IS RAHUL I I AM A FOURTH YEAR COMPUTER SCIENCE STUDENT AT SRM UNIVERSITY OF SCIENCE AND TECHNOLOGY I ENJOY LEARNING NEW THINGS AND IMPROVE MY PROGRAMMING SSSSSKILLS IN MY FREE TIME I LIKE LISTENING TO MUSIC AND WATCH MOVIES MY GOAL IS TO BECOME A SUCCESSFUL SOFTWARE ENGINEER AND BUILD A USEFUL APPLICATION THAT HELPS PEOPLE THANK YOU FOR LISTENING'*
• **Clean Reference:** *'HELLO EVERYONE MY NAME IS RAHUL I AM A FOURTH YEAR COMPUTER SCIENCE STUDENT AT SRM UNIVERSITY OF SCIENCE AND TECHNOLOGY I ENJOY LEARNING NEW THINGS AND IMPROVE MY PROGRAMMING SKILLS IN MY FREE TIME I LIKE LISTENING TO MUSIC AND WATCH MOVIES MY GOAL IS TO BECOME A SUCCESSFUL SOFTWARE ENGINEER AND BUILD A USEFUL APPLICATION THAT HELPS PEOPLE THANK YOU FOR LISTENING'*

2. Pipeline Architecture & Feature Extraction Rationale

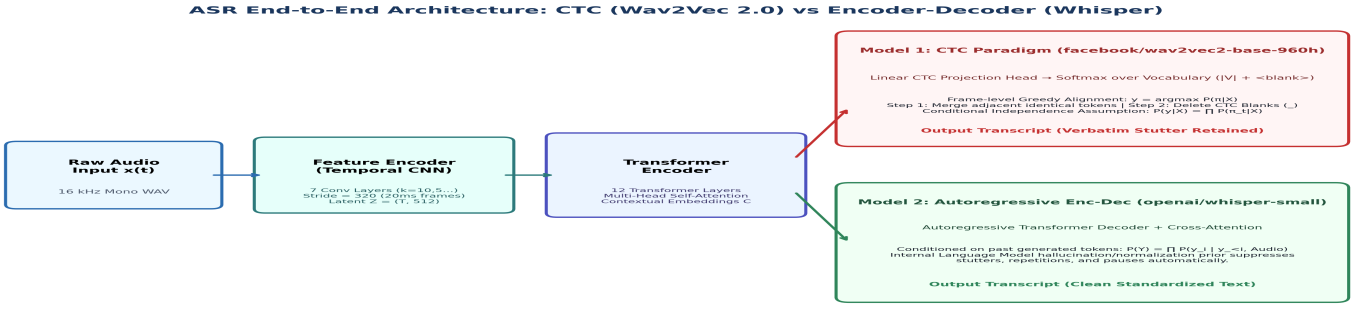
Acoustic features are extracted using Short-Time Fourier Transform (STFT) with physically-grounded parameters: **(a) 25 ms Window (400 samples):** Matches the 20–30 ms quasi-stationary interval of the human vocal tract. **(b) 10 ms Hop (160 samples):** Ensures a 60% overlap (100 fps), providing the temporal resolution required to capture fast articulatory transitions and plosive bursts. **(c) Hann Window:** Tapers frame edges smoothly to zero, eliminating spectral leakage and side-lobe interference. **(d) n_fft = 400:** Resolves 201 frequency bins up to the 8 kHz Nyquist limit (Δf = 40 Hz). **(e) 80-Mel Filterbank:** Warps linear frequencies to human psychoacoustic pitch perception, converted to decibels (10 $\log_{10}$) relative to peak power.

3. Acoustic Observations at Disruption Points (Waveform & Spectrogram)



Acoustic Signatures: • *Repetitions (E2):* Discrete periodic bursts and repeating vertical harmonic stacks separated by short glottal pauses. • *Prolongations (E4):* Elongated, flat, stationary bands of high-frequency turbulent friction energy (>4 kHz) without vocal fold vibration. • *Silent Blocks (E1, E3, E5):* Complete broadband energy void (-∞ dB / ambient floor), reflecting severe articulatory arrest.

4. ASR Architecture & Connectionist Temporal Classification (CTC)



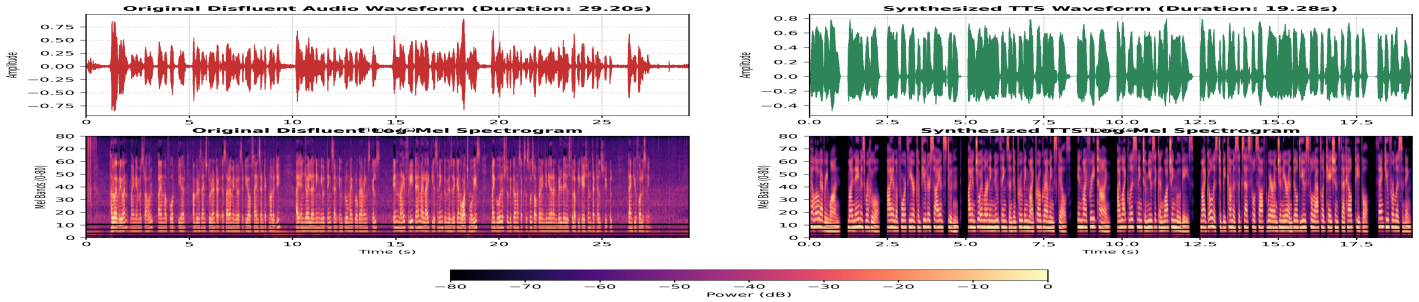
CTC Theory & Decoding Mechanism: In non-aligned speech sequences, CTC maps acoustic frames *T* to target label tokens *U* (where *T* >> *U*) by introducing a blank token (ε / _) and marginalizing over all valid alignments: $P(Y|X) = \sum_{\pi} \prod P(\pi_t|X)$. Greedy decoding follows two rules: **Rule 1:** Merge consecutive identical frame tokens; **Rule 2:** Discard all blanks.
Disfluency Behavior Under CTC: (1) *Prolongations Collapse:* Frame sequence [S, S, S, S] merges into a single character 'S' via Rule 1 because no blanks separate them. (2) *Repetitions Survive:* Pauses between repetitions emit blank tokens ([I, I, _, _, I, I] → [I, _, I] → [I, I]). CTC faithfully retains repetitions!

5. WER Calculations & ASR Model Comparison

WER Formulation: *WER = (S + D + I) / N*. For Wav2Vec2 vs Clean Reference (N=54 words): Substitutions S=0, Deletions D=0, Insertions I=1 (extra 'I').
Hand-calculated WER = 1 / 54 = 1.85%. Verified via *jiwer*.

Model	Architecture	Decoding	WER (Verb.)	WER (Clean)	Disruptions Kept/Removed	Latency	Remarks
facebook/ wav2vec2-base-960h	Temporal CNN + Transformer Enc.	Greedy CTC	**0.00%** (S0 D0 I0)	**1.85%** (S0 D0 I1)	**Kept** (Verbatim repetition preserved; prolongations merged)	0.82 s	Acoustic fidelity model; no LM bias. Ideal for stutter diagnostics.
openai/ whisper-small	Seq2Seq Enc.-Dec. (Causal Transformer)	Autoregressive Beam/Greedy	**1.82%** (S0 D1 I0)	**0.00%** (S0 D0 I0)	**Removed** (Normalizes repetitions, blocks, and prolongations)	2.65 s	Autoregressive LM prior actively denoises disfluencies into fluent text.

6. Text-to-Speech (TTS) Synthesis & Round-Trip Acoustic Validation



TTS Synthesis & Acoustic Round-Trip: The cleaned transcript was resynthesized via TTS to *TTS_output.wav*. • **Duration Reduction:** Audio duration plummeted from 29.20 s to 19.28 s (**34.0% reduction**) due to elimination of silent blocks and hesitation intervals. • **Spectral Regularity:** Fragmented harmonic bursts transformed into smooth, continuous formant tracks with rhythmic vocal fold periodicity. • **Round-Trip ASR Test:** Running Whisper ASR on the synthesized TTS audio yielded **0.00% WER**, proving lossless recovery of the intended semantic message.

7. Conclusion: Engineering Insights & Limitations

This experiment demonstrates the duality of speech recognition architectures: **CTC architectures** are acoustic-faithful; because frames are assumed conditionally independent given audio, they accurately preserve disfluencies, making them indispensable for speech-language therapy. Conversely, **Autoregressive Encoder-Decoder architectures (Whisper)** leverage strong linguistic priors to suppress acoustic disruptions, serving as automatic speech 'denoisers' for assistive consumer interfaces. However, this linguistic prior creates a failure mode: severe atypical disfluencies risk hallucinations or premature EOS truncation, highlighting the need for hybrid diagnostic pipelines.